



C. V. Raman
Global University

DATA COMMUNICATION & COMPUTER NETWORK (DCCN)

[CS221] [Cr:3]



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Syllabus

Unit-1: Data Communication Fundamentals

[8 Hrs]

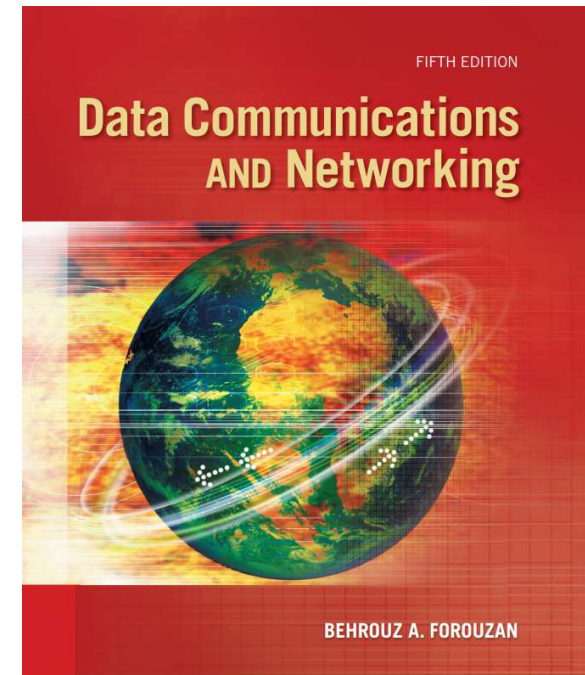
Overview of Data Communications, Fundamentals of Network, Network Topologies, Services, Standardization, Reference Models: OSI Model, TCP/IP Model.

Physical Layer: Analog and Digital Signals, Data Rate Limits, Transmission Impairment, More about signals. Digital Transmission: Line coding, Sampling, Transmission modes.

Unit-2: Modulation and Switching Techniques

[8 Hrs]

Analog Transmission: Modulation of Digital Data, Modulation of Analog signals. Multiplexing: FDM, WDM, TDM, Transmission Media: Guided Media, Unguided Media, Circuit Switching and Packet Switching.



Syllabus

Unit-3: Data Link Layer

[8 Hrs]

Error Detection and Correction: Types of Errors, Detection, Error Correction.

Data Link Control and Protocols: Flow and Error Control, Stop-and-wait ARQ, Go-Back-N ARQ, Selective Repeat ARQ, HDLC, PPP.

Multiple Accesses: Random Access, Controlled Access, Channelization. Local area Network: Standard Ethernet, Wireless LANs: IEEE 802.11.

Unit-4: Network and Transport Layer:

[6 Hrs]

Host to Host Delivery: Internetworking, Addressing and Routing, Network Layer Protocols: ARP, IPV4, ICMP, IPV6,

Transport Layer: Process to Process Delivery: UDP, TCP, Congestion Control. Routing Protocols: AODV, DSDV, DSR.

Unit-5: Application Layer:

[6 Hrs]

Client Server Model, Socket Interface, Domain Name System (DNS): Electronic Mail (SMTP), File Transfer (FTP), POP, Remote Logging, HTTP and WWW.

Text / References Book

Text Books

T1: Data Communications and Networking: Behrouz A. Forouzan, Tata McGraw-Hill, 4th Edition.

T2: Computer Networks: A. S. Tannenbum, D. Wetherall, Prentice Hall, Imprint of Pearson 5th Edition.

References Books

R1: Computer Architecture and Parallel Processing”, Hwang and Briggs, Tata McGraw Hill Publication,

R2: Data and Computer Communications: William Stallings, Prentice Hall, 9th Edition.

Websites References

1. <https://www.youtube.com/watch?v=sG6WGvzmVaw>
2. <https://www.javatpoint.com/computer-network-introduction>
3. <https://www.geeksforgeeks.org/basics-computer-networking/>
4. https://www.tutorialspoint.com/computer_fundamentals/computer_networking.htm
5. <https://nptel.ac.in/courses/106/105/106105081/>
6. <https://www.classcentral.com/course/swayam-computer-networks-13951>
7. <https://www.coursera.org/courses?query=computer%20network>

Course outcomes

CO1	Understand the fundamental principles of Data communication and Computer Networking.
CO2	Analyze different features of analog and digital transmission.
CO3	Analyze the working principles and protocols of data link layer.
CO4	Identify and differentiate working principles and protocols of network and transport layer.
CO5	Identify and implement different types of application in application layer

Data Communication and Computer Networks

DATA COMMUNICATIONS

- The term **telecommunication** means communication at a distance.
- The word **data** refers to information presented in whatever form is agreed upon by the parties creating and using the data.
- **Data communications** are the exchange of data between two devices via some form of transmission medium such as a wire cable.

Fundamental Characteristics:

Delivery, Accuracy, Timeliness, Jitter

Fundamental Characteristics

Effectiveness of data communication system depends on

Delivery: The system must deliver data to correct destination.

Data received by the intended user only

Accuracy: The system must deliver data accurately (no change).

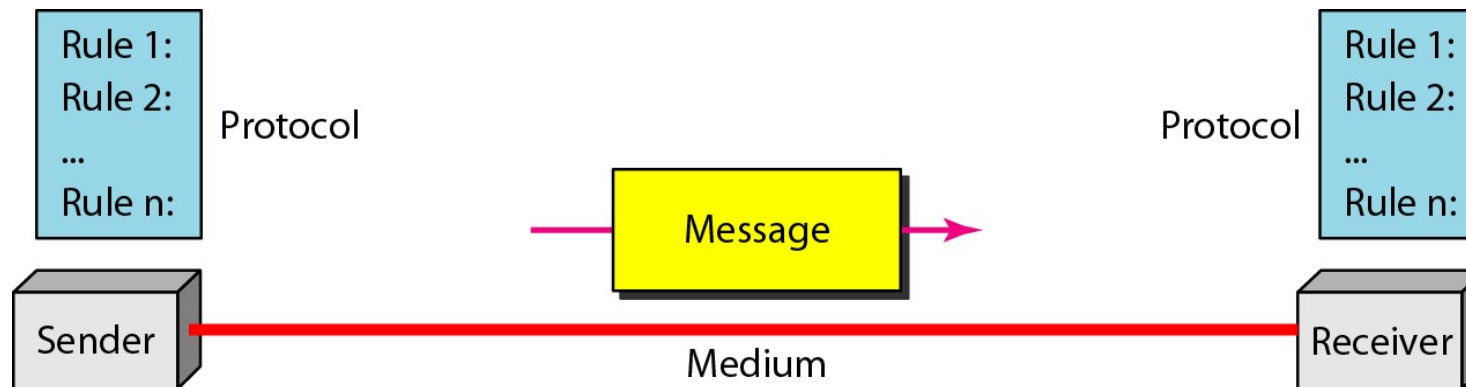
Data changed & uncorrected is unusable

Timeliness: The system must deliver data in timely manner

- Data arrived late are useless
- In the same order (video and audio) & without delay (Real time transmission)

Jitter: Variation in the packet arrival time (uneven quality in the video is the result)

Five components of data communication



Components

1. Message,
2. Sender,
3. Receiver
4. Transmission Medium
5. Protocol

Components of Data Communication

Message: The **message** is the information (**data**) to be communicated.

Examples: Text, numbers, pictures, audio, and video

Sender: The **sender** is the device that sends the data message.

Examples: Computer, workstation, telephone handset, video camera, and so on

Receiver: The **receiver** is the device that receives the message.

Examples: Computer, workstation, telephone handset, television, and so on

Medium: The **transmission medium** is the physical path by which message travels from sender to receiver.

Examples: Twisted-pair wire, coaxial cable, fiber-optic cable and radio waves.

Protocol: Set of rules that governs data communications. It represents an agreement between the communicating devices. Without a protocol, two devices may be connected but not communicating.

Data Flow

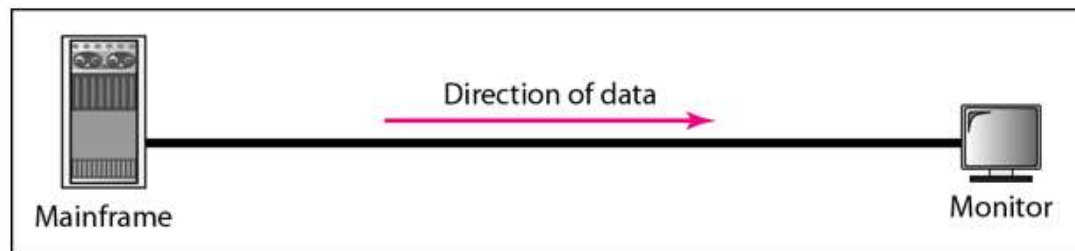
Communication between two devices can be

- simplex,
- half-duplex,
- full-duplex

Simplex Mode

- In **simplex mode**, the communication is unidirectional. Only one of the two devices on a link can transmit; the other can only receive.
- The simplex mode can use the entire capacity of the channel to send data in one direction.

Examples: Keyboards and traditional monitors.

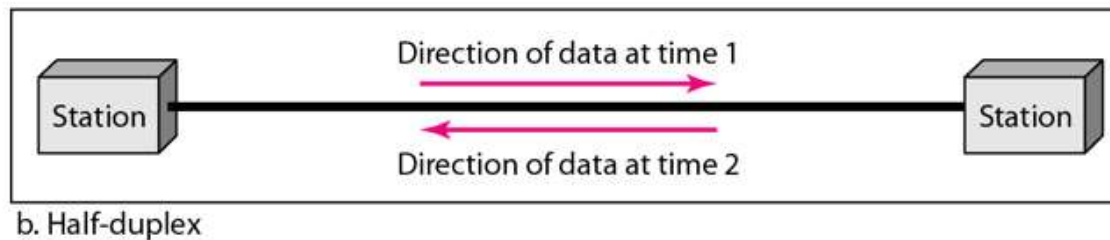


a. Simplex

Half-Duplex Mode

- In **half-duplex mode**, each station can both transmit and receive, but not at the same time.
- When one device is sending, the other can only receive, and vice versa.
- The entire capacity of a channel is taken over by the transmitting device

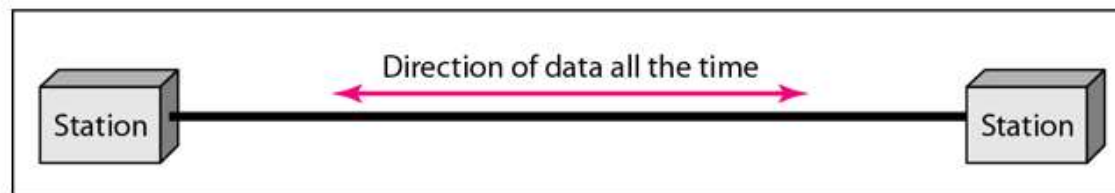
Example: Walkie-talkies



Full-Duplex Mode

- In **full-duplex mode** (also called *duplex*), both stations can transmit and receive simultaneously
- The capacity of the channel is divided between signals traveling in both directions

Example: Telephone network



c. Full-duplex

NETWORKS

A **network** is a set of devices (often referred to as **nodes**) connected by communication **links**.

Node: A node can be a computer, printer, or any other device capable of sending and/or receiving data generated by other nodes on the network.

Links: Communication pathway that transfers data from one device to another.
Ex: Cable, air, optical fiber etc

Topics discussed in this section:

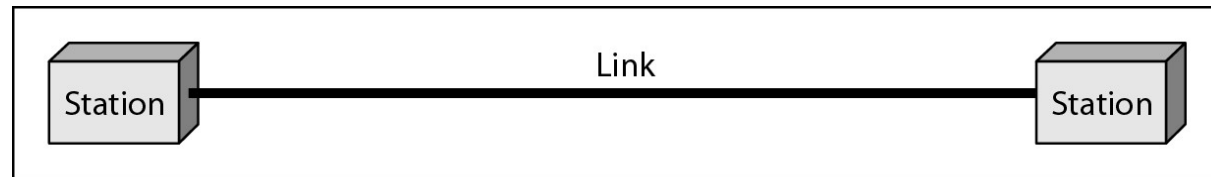
- **Network Criteria**
- **Physical Structures**
- **Categories of Networks**

Network Criteria

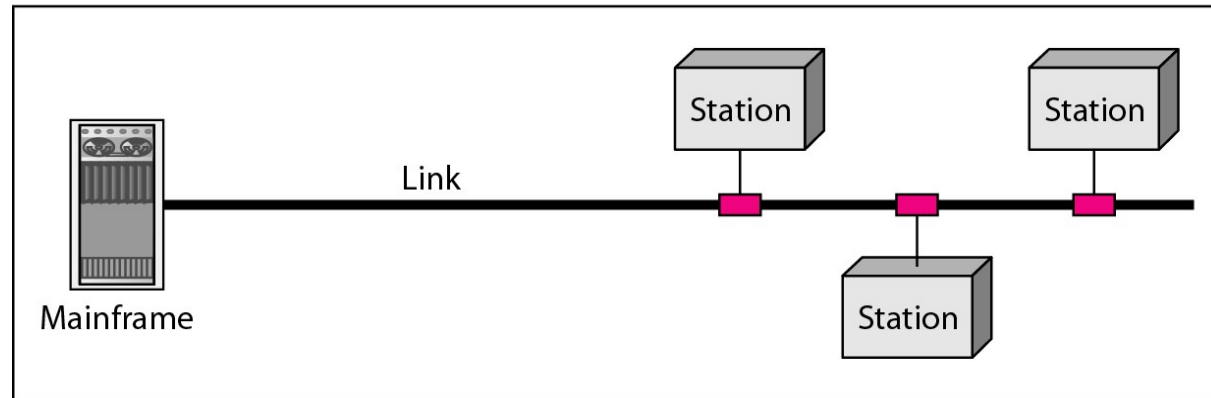
- **Performance** can be measured in many ways, including transit time and response time.
 - Throughput
 - Delay
- **Reliability** is measured by the frequency of failure, recovery, and robustness.
- **Security** issues include protecting data from unauthorized access.

Types of connections: point-to-point and multipoint

- A **point-to-point** connection provides a dedicated link between two devices.
- A **multipoint connection** is one in which more than two specific devices share a single link.
- In a multipoint environment, the capacity of the channel is shared, either spatially or temporally.
- If several devices can use the link simultaneously, it is a *spatially shared* connection. If users must take turns, it is a *timeshared* connection.



a. Point-to-point

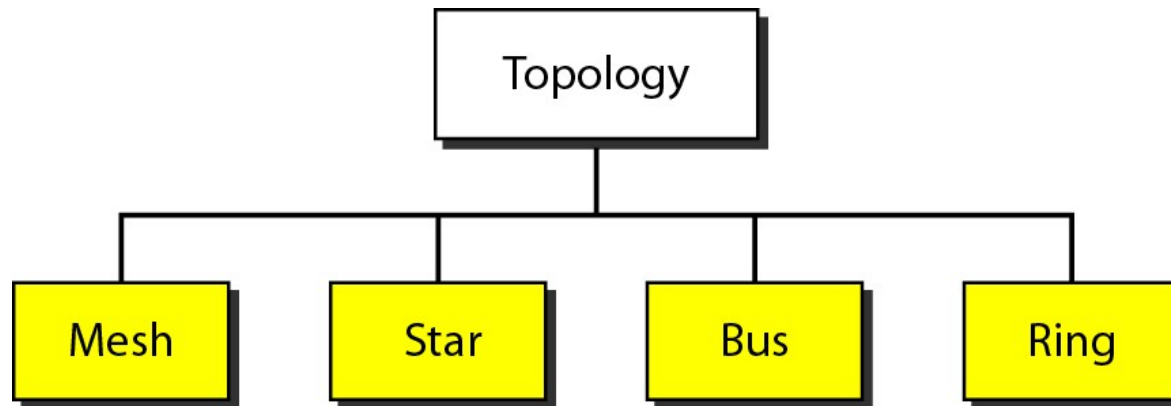


b. Multipoint

Physical Topology

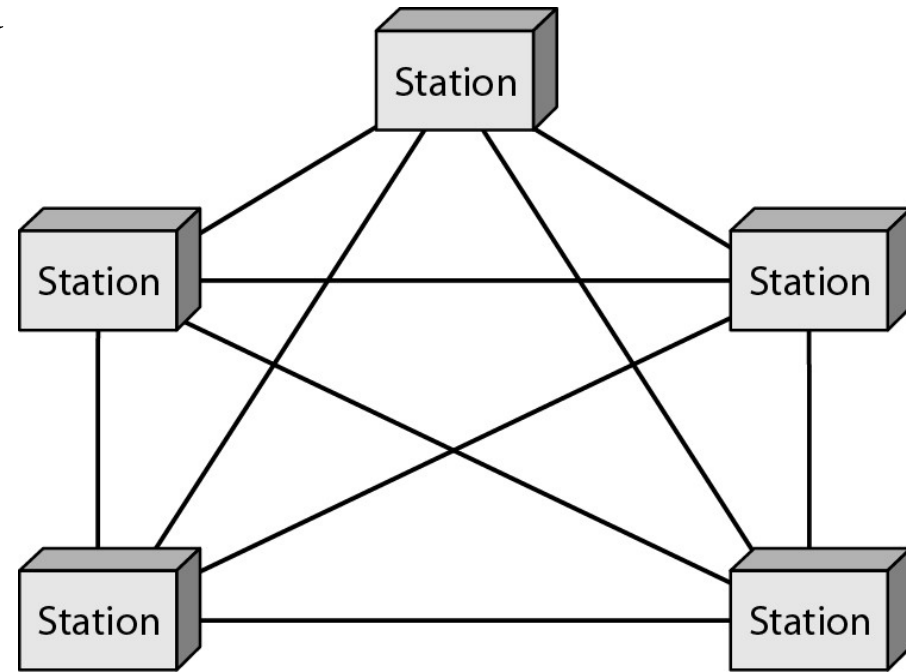
The topology of a network is the *geometric representation* of the relationship of all the *links* and linking devices (usually called *nodes*) to one another.

There are four basic topologies possible: mesh, star, bus, and ring.



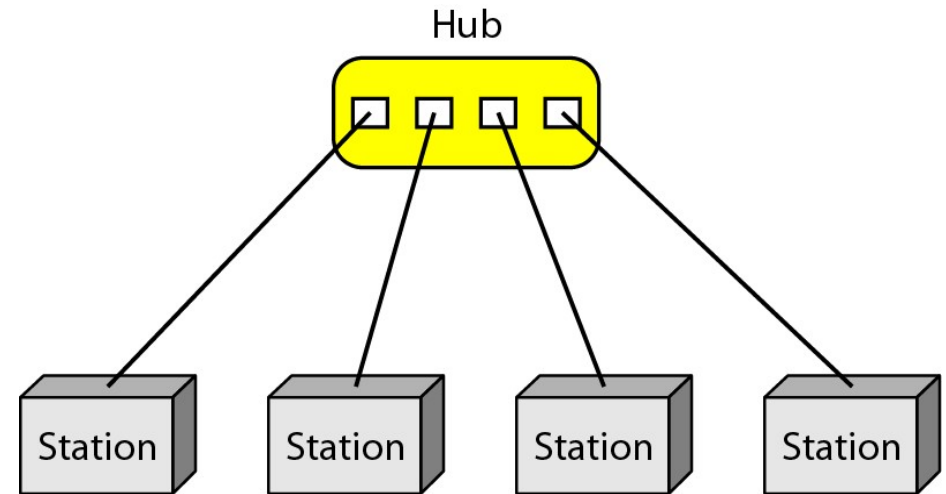
Mesh Topology

- In a **mesh topology**, every device has a dedicated point-to-point link to every other device.
 - # Cables = $n(n - 1) / 2$ (No. of duplex mode)
 - # ports = $n(n - 1)$
- **Advantages:**
 - Minimum traffic problems
 - Robustness
 - Security
 - Easy fault identification
- **Disadvantages:** Amount of cabling and the number of I/O ports required.
- **An application:** connection of telephone regional offices.



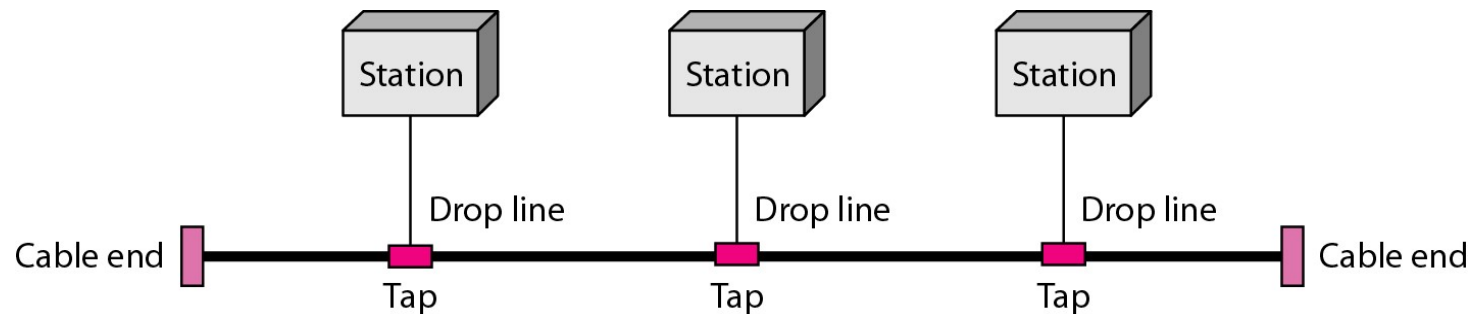
Star Topology

- In a **star topology**, each device has a dedicated point-to-point link only to a central controller, called **hub**.
- Controller relays the data to the other connected device.
- **Advantages:**
 - Robustness
 - Easy fault identification
- **Disadvantages:** Single point failure.
- **An application:** Local-area networks (LANs).



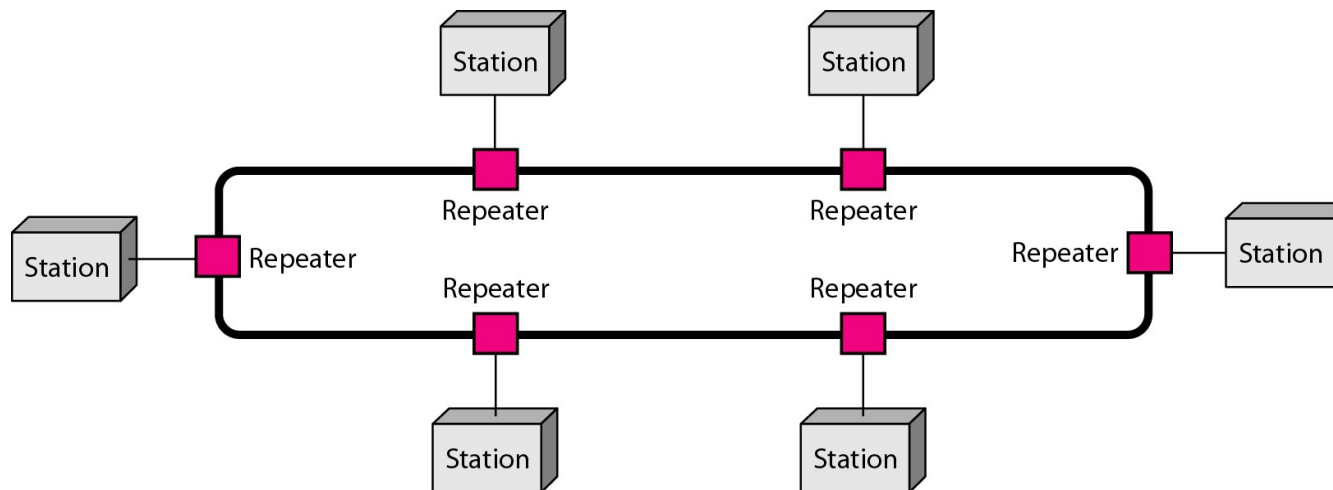
Bus Topology

- A **bus topology** is a multipoint topology. One long cable acts as a **backbone**.
- **Advantages:**
 - Less cabling than mesh or star topologies.
- **Disadvantages:**
 - Difficult reconnection.
 - Signal reflection at the taps can cause degradation in quality.
 - Single point failure.
- **An application:** Earlier Local-area networks (LANs).

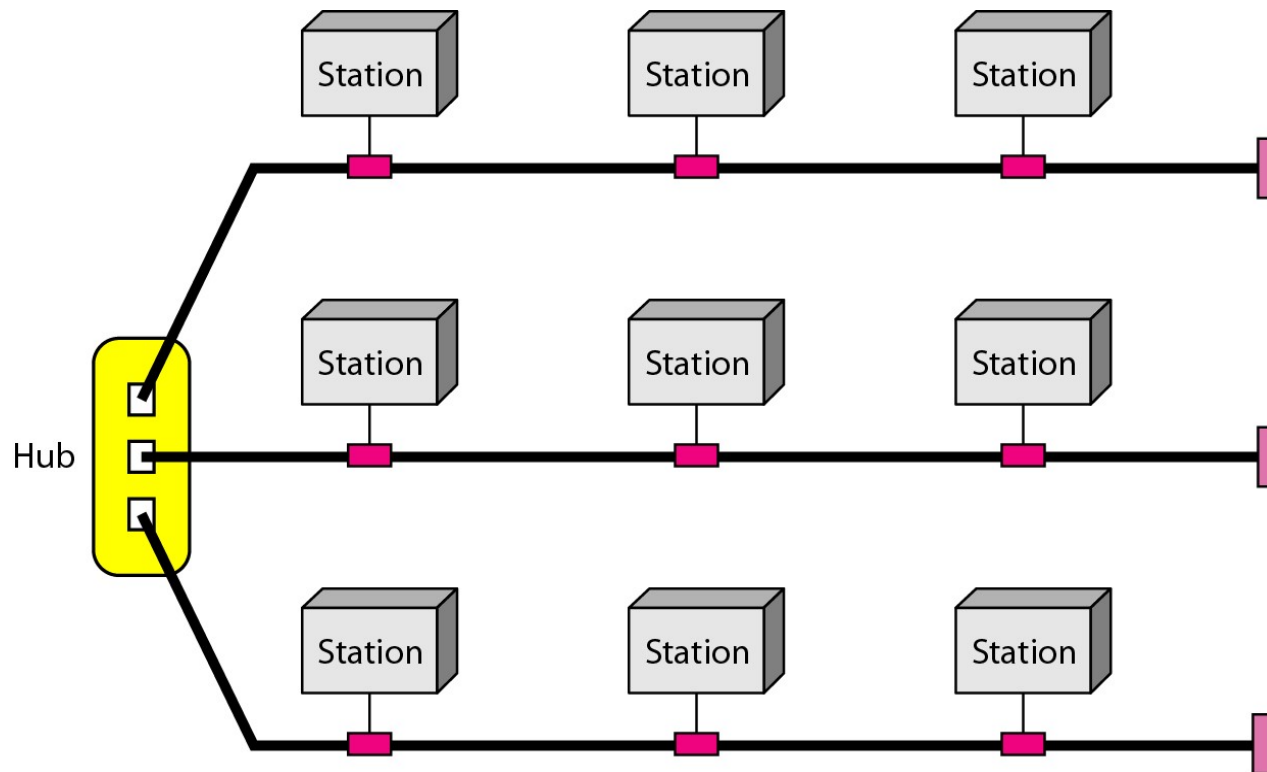


Ring Topology

- In a **ring topology**, each device has a dedicated point-to-point connection with only the two devices on either side of it.
- Each device incorporates a repeater.
- **Advantages:**
 - Easy to add device.
- **Disadvantages:**
 - Unidirectional traffic.
- **An application:** Earlier LAN of IBM.

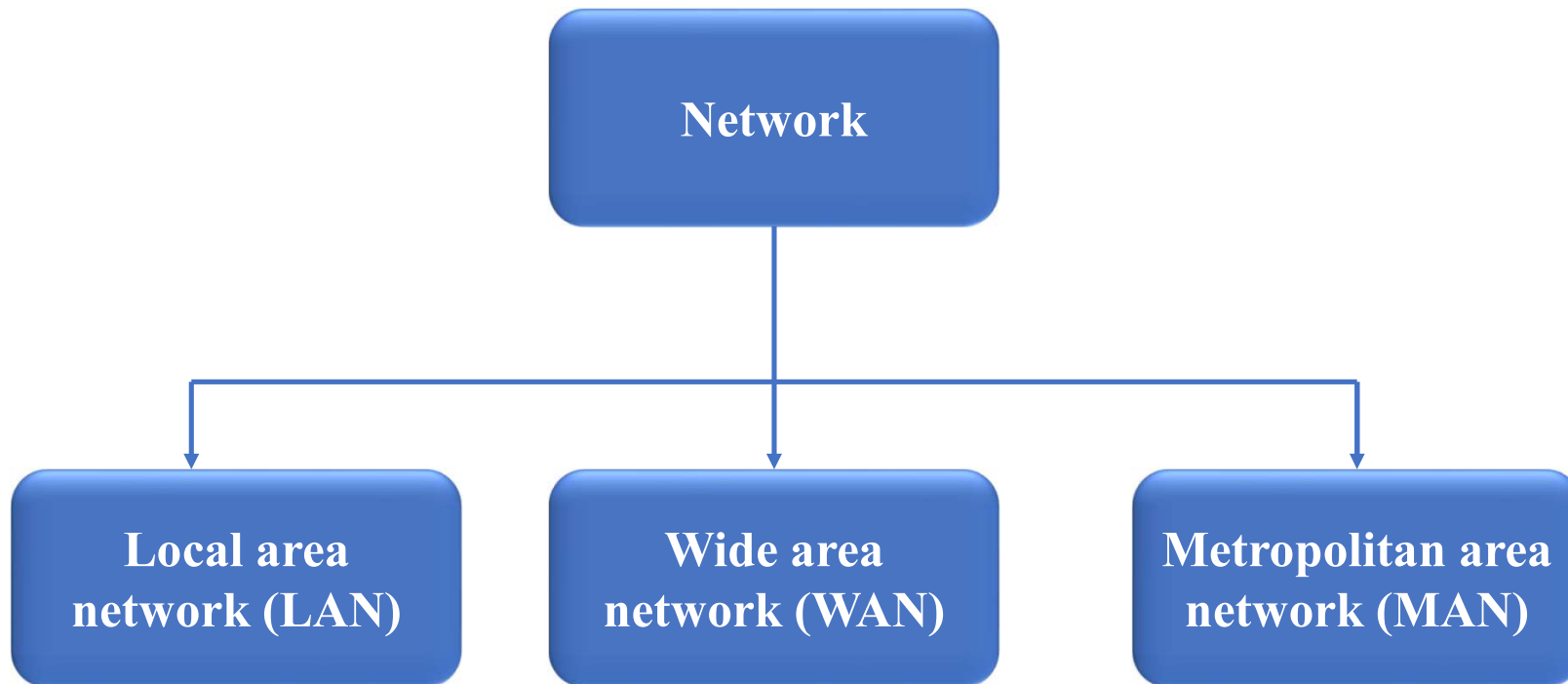


A hybrid topology: a star backbone with three bus networks



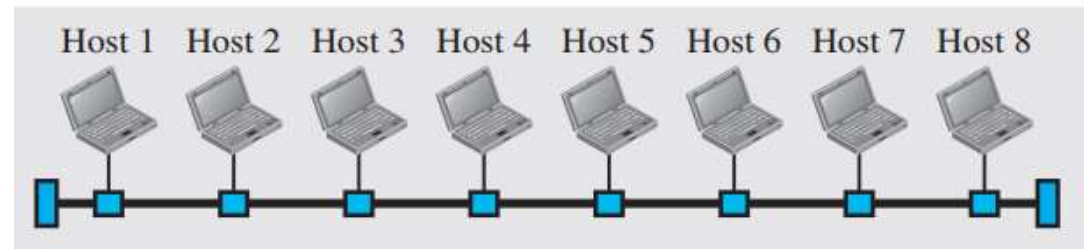
Network Categories

Network category is determined by its size and its physical architecture

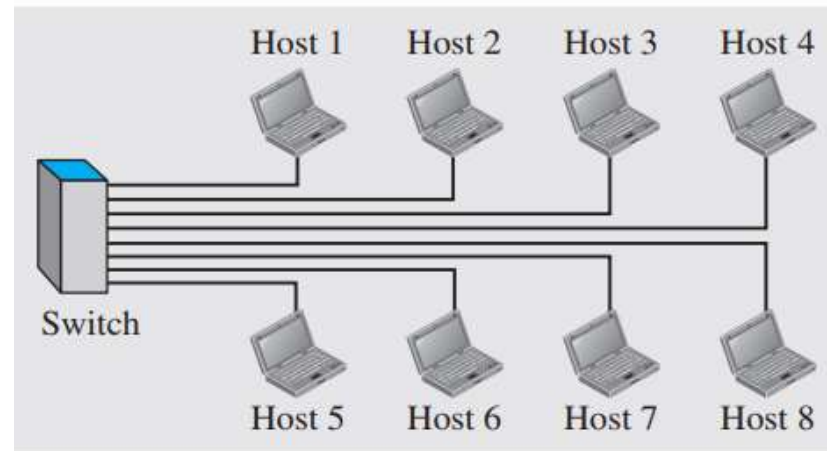


Local area network (LAN)

- A **local area network (LAN)** is usually privately owned and connects some hosts in a single office, building, or campus (like CGU).
- Each host in a LAN has an address.
- Definition of a LAN does not define the minimum or maximum number of hosts in a LAN.
- LAN use any physical topology (**Bus**, **ring** and **star**).
- LAN covers an area < 2 miles.



a. LAN with a common cable (past)



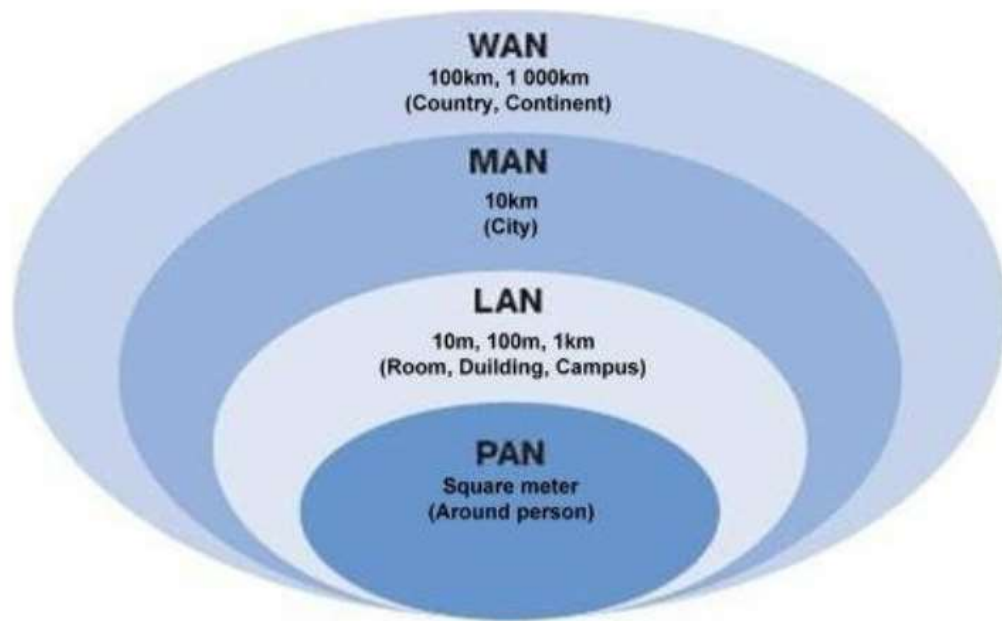
b. LAN with a switch (today)

Legend

-  A host (of any type)
-  A switch
-  A cable tap
-  A cable end
-  The common cable
-  A connection

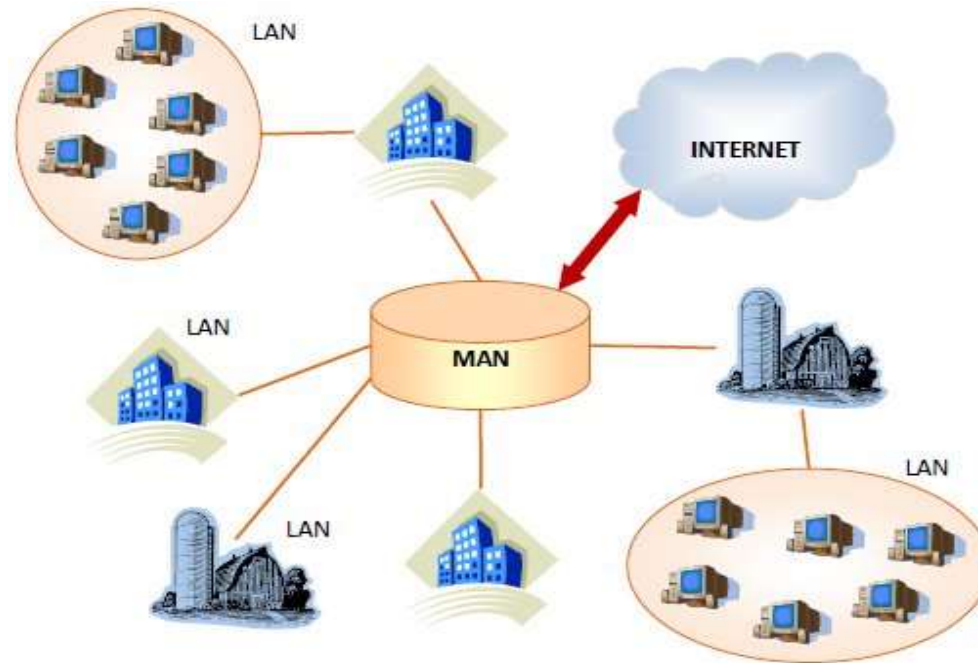
Wide area network (WAN)

- Provide long distance transmission of data, image, audio and video information over large areas.
- Covers a large geographical area, like a country or continent. Connects LANs and MANs.



Metropolitan area network (MAN)

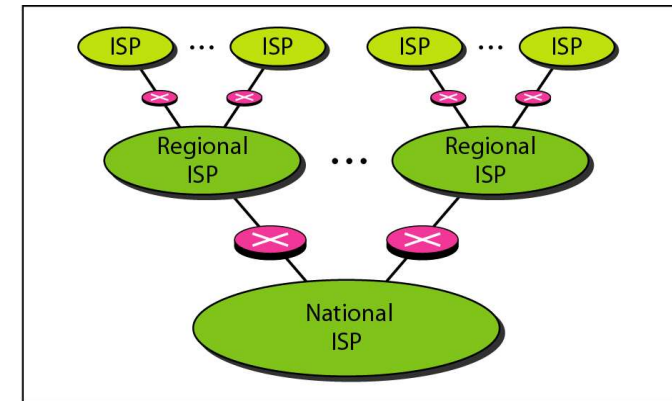
- Its size is in between LAN and WAN.
- Covers the area inside a town or a city.
- Example: Cable TV network used for high-speed data connection to the internet.



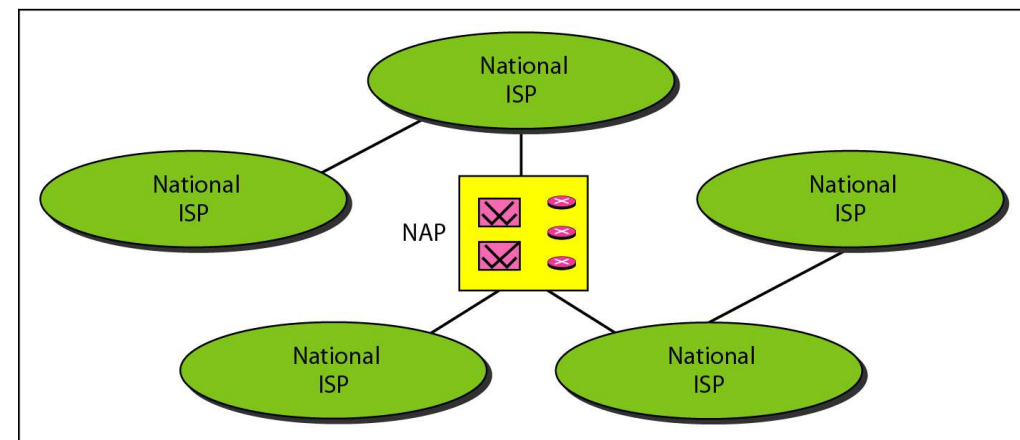
Parameter	LAN (Local Area Network)	MAN (Metropolitan Area Network)	WAN (Wide Area Network)
Coverage Area	Covers a small geographical area such as a room, building, office, or campus.	Covers a city or metropolitan area.	Covers a country, continent, or the entire world.
Ownership	Privately owned and managed by a single organization or institution.	Owned and managed by a government, ISP, or large organization.	Managed by telecom operators, ISPs, or multiple organizations.
Transmission Speed	Very high (typically 100 Mbps to several Gbps).	High, but generally lower than LAN.	Lower than LAN and MAN due to long-distance communication.
Installation Cost	Low	Moderate	High
Transmission Medium	Ethernet cables, Fiber Optic cables, and Wi-Fi.	Fiber Optic cables, and Microwave links.	Fiber Optic, Leased Lines, Satellite links
Propagation Delay	Very low because of the short communication distance.	Moderate due to wider geographical coverage.	High because data travels over long distances.
Examples	Home network, office network, computer laboratory, college campus network.	City-wide university network, cable TV network, smart city network	Internet, banking network,, multinational corporate network.

Internet Today

- Made of many LANs and WANs
- Every day new networks area added and removed
- Internet services Providers (ISPs) offer services to the end users
 - International service providers
 - National service providers
 - Regional service providers
 - Local service providers



a. Structure of a national ISP



b. Interconnection of national ISPs

Internet Today: Examples

Level	Coverage Area	Examples	Connection to Other Levels
Local ISP	Specific city/town or small region	Ortel (Bhubaneswar), Wishnet (Kolkata)	Often franchisees or resellers of regional/national providers (e.g., RailWire LCOs)
Regional ISP	One or a few states or zones	Alliance Broadband (WB, Odisha), ACT (South India)	May partner with national backbones; could expand to national level
National ISP	Entire country	BSNL, Jio, Airtel	Provide backbone connectivity; may resell bandwidth to regional/local ISPs
International ISP/Carrier	Multiple countries across continents	Tata Communications, Airtel Business, Orange	Own/manage global submarine cables; provide bandwidth to national ISPs

Protocols

A protocol is a set of rules that govern data communications. Defines the methods of Communication, what to communicate, when to communicate, how to communicate etc.

1. **Syntax**: Syntax means format of data or the structure how it is pretend. For example,

- ✓ First eight bits for sender address
- ✓ next eight bits for receiver address
- ✓ rest of the bits for message/information.

1	1	0	0	1	1	0	1	0	1	1	0	1	1	1	0	1	1	1	0	0	0	1	0	1	1	1	1	0	.	.	.
---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---	---

2. **Semantics**: Semantics is the meaning of each section of bits e.g. The address bit means the route of transmission or final destination of the message.

3. **Timing**: Timing means, when data should be sent and how fast they can be sent

Example: Sender produces data at 100 Mbps but the receiver can process data at only 1 Mbps

⇒ overload and data will be lost.

Standards

- Standard **provides guidelines** to manufacturers, vendors, government agencies, and other service providers. It **ensure** about the **interconnectivity and compatible** of the devices.
- It helps in creating and maintaining an open and competitive market for equipment manufactures.

Data communication standards fall into two categories:

1. De facto

- ✓ De facto means by facts or by conventions.
- ✓ The standards that are not approved by an organized but are widely used are De facto standards. These are established by manufacturer.

2. De jure:

- ✓ De jure means by law or by regulations.
- ✓ These are the standards that are recognized officially by organizations.

Standards

- ❑ International Standards Organization (ISO): Developing cooperation in the realms of scientific, technological and economic activity.
- ❑ International Telecommunication Union (ITU) : Research and establishment of standards for telecommunication in general
- ❑ Institute of Electrical and Electronics Engineers (IEEE): It aims to advance theory, creativity and product quality in the fields of electrical engineering, electronics and radio as well as in all related branches of engineering.
- ❑ Electronic Industries Association (EIA): Promotion of electronics manufacturing concerns.